FraudLens

AI-Powered Mule Account & Suspicious Transaction Detection System

CyberShield PSBs Hackathon Series 2026

Bank of India + IIT Hyderabad | Prize Pool: Rs. 20 Lakhs

An Initiative of Govt. of India, Ministry of Finance, Dept. of Financial Services

Team Leader	**Ashutosh**
Member 2	**Diksha**
Member 3	**Divyansh Pratap Singh**
Member 4	**Garvit Agrawal**
Institution	**GL Bajaj Group of Institutions, Mathura**
Problem Statement	**PS-2: Mule Account & Suspicious Transaction Detection**
GitHub Repository	https://github.com/dikshaashadeep-ship-it/fraudlens-hackathon

Registration Deadline: 15 June 2026 | Grand Finale: 27–28 August 2026

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1. Problem Statement

Every day, criminals move stolen money through innocent people's bank accounts — these are called **mule accounts**. Detecting them manually from millions of daily transactions is impossible for investigators. Traditional rule-based systems produce excessive false alerts and miss hidden fraud networks spanning multiple accounts.

Our solution: An AI-powered system that automatically detects suspicious transactions, discovers mule account networks, assigns risk scores, and explains each fraud case in plain English — all from a simple CSV file upload on a clean web dashboard.

2. Our Unique Idea — FraudLens

> *"An AI-powered fraud intelligence platform that detects suspicious transactions, uncovers mule-account networks, and generates explainable risk insights in real time."*

Feature 1 — Visual Mule Network Graph

Every bank account is drawn as a node and every money transfer as an edge. Mule accounts appear as heavily connected hub-nodes — investigators can *see* the fraud network at a glance instead of reading thousands of rows.

Feature 2 — Gemini AI "Why is this fraud?" Explainer

For every flagged account, the system calls Gemini API and auto-generates a 3-line plain-English explanation. No other beginner team will have AI-generated explanations — this is our key differentiator.

> *"Account A1234 received Rs. 80,000 from 6 unrelated sources in 2 hours and transferred everything out within 15 minutes — mule account behavior. Risk Score: 91/100."*

Feature 3 — Fraud Alert Timeline

Flagged events appear in a live timeline sorted by time and risk score — like a news feed of fraud alerts. Intuitive and actionable for investigators.

3. Risk Scoring Model

Every transaction receives a score out of 100 based on weighted indicators:

Fraud Indicator	Points	Risk Band	Action
Large Transaction Amount	+30		Unusual spending pattern
Rapid Multiple Transfers	+20		Quick fund movement
Mule Account Pattern	+40		Relay behavior detected
Blacklisted Receiver	+50		Known fraud account
Score 0 – 30	**Low**	■	**Normal — monitor only**
Score 31 – 70	**Medium**	■	**Flag for review**
Score 71 – 100	**High**	■	**Immediate action needed**

4. System Workflow

CSV upload → data validation & cleaning → rule-based risk scoring → ML anomaly detection → graph-based mule discovery → Gemini AI explanation → investigator dashboard with alerts & downloadable report.

5. Technology Stack

Tool	Used For	Why Chosen
Python + Pandas	Data loading, cleaning, analysis	Easy to learn, most used in data science
Scikit-Learn	ML fraud detection model	Free, beginner-friendly, powerful
NetworkX + Plotly	Transaction graph + visualization	Industry standard for fraud graph analytics
Streamlit	Web dashboard + CSV upload	Build web app in pure Python — no HTML needed
Gemini API (Free tier)	AI fraud explanations	Best-in-class GenAI, free for students
GitHub	Team collaboration	Version control + submission

6. Team Roles & Responsibilities

Member	Role	Key Responsibility
Ashutosh	Team Lead + ML Eng.	Dataset prep, fraud detection model, risk scoring
Diksha	Dashboard Developer	Streamlit UI, Plotly charts, Gemini API integration, PPT
Divyansh P.S.	Graph Analytics Eng.	NetworkX transaction graph, mule cluster detection
Garvit A.	Data + Integration	Data pipeline, module integration, testing, documentation

7. Project Timeline

Phase	Deliverable
Week 1 (Now – 10 June)	Register, form team, GitHub repo, sample CSV dataset
Week 2 (10 – 17 June)	Pandas cleaning, risk scoring logic, basic Streamlit dashboard
Week 3 (17 – 24 June)	ML model (Isolation Forest), NetworkX graph, module integration
Week 4 (24 June – 1 July)	Gemini API explanations, fraud timeline, UI polish
Week 5–7 (July – 15 Aug)	Testing, bug fixes, PPT + architecture diagram, demo video
Final Week (before 27 Aug)	Full rehearsal, final testing, IIT Hyderabad Grand Finale

8. Dataset Description

Since real bank transaction data is confidential, we use a **synthetic (AI-generated) dataset** that accurately mimics real-world fraud patterns — standard practice in academic and hackathon fraud-detection projects.

Dataset 1 — Synthetic Transaction Dataset (Primary)

Generated using Python (Pandas + NumPy). 500–1000 rows of realistic bank transactions.

Column	Type	Description	Example
transaction_id	String	Unique transaction ID	TXN00123
sender_account	String	Account sending money	ACC0045
receiver_account	String	Account receiving money	MULE002
amount	Float	Transaction amount (Rs.)	82500.00
timestamp	DateTime	Date and time of transaction	2024-01-15 14:32
location	String	City of transaction origin	Mumbai
is_blacklisted	Integer	1 if receiver is blacklisted, else 0	1
risk_score	Integer	Fraud risk score (0–100)	91

Dataset 2 — Fraud Alerts Dataset (Secondary)

Simulates government cyber-fraud tickets and bank fraud monitoring alerts.

Column	Type	Description	Example
alert_id	String	Unique alert ID	ALT00056
account_id	String	Flagged account	MULE002
alert_type	String	Type of fraud alert	Rapid Transfer
severity	String	Low / Medium / High	High
reported_by	String	Bank System or Govt Cyber Cell	Govt Cyber Cell
alert_date	Date	When alert was generated	2024-01-15

Dataset Summary

Aspect	Detail
Normal transactions	75% of data — amounts Rs.500–25,000, varied accounts, normal timing
Mule transactions	25% of data — large amounts Rs.40,000–1,50,000, rapid transfers within 30 min
Mule accounts	5 accounts (MULE001–MULE005) acting as fund relay hubs
Blacklisted accounts	3 known bad accounts pre-flagged in the dataset
Additional dataset	Kaggle IEEE-CIS Fraud Detection dataset for ML model validation
Reproducibility	Python random seed=42 — generation script on GitHub repository

9. Detailed Solution Approach

Step 1 — Data Ingestion & Cleaning

User uploads transaction CSV via Streamlit. Python/Pandas handles missing values, data type validation, timestamp parsing, and feature engineering (e.g. transaction frequency per account per hour).

Step 2 — Rule-Based Risk Scoring

Each transaction is scored 0–100 using weighted indicators: large amount (+30), rapid transfers (+20), mule relay pattern (+40), blacklisted receiver (+50). Scores above 70 are flagged High Risk.

Step 3 — ML Anomaly Detection (Isolation Forest)

Scikit-Learn Isolation Forest detects statistical anomalies that rule-based scoring may miss. Anomaly prediction is combined with the risk score for final classification.

Step 4 — Graph-Based Mule Detection (NetworkX)

Accounts modelled as graph nodes; transactions as directed edges. High in-degree (many senders) + high out-degree (many receivers) = probable mule node. Community detection reveals hidden fraud clusters.

Step 5 — Gemini AI Explanation Engine

For each High Risk account, a structured prompt is sent to Gemini API. Response is a 3-line plain-English investigation summary — immediately actionable for non-technical investigators.

Step 6 — Interactive Streamlit Dashboard

All outputs on one web app: color-coded risk table, Plotly charts, NetworkX network graph, fraud alert timeline, Gemini explanation panel, downloadable report.

10. Expected Impact

Without FraudLens	With FraudLens
Investigators manually review thousands of rows	AI auto-flags top suspicious accounts — reduces review time by ~80%
Days to detect a mule network	Mule network detected in under 5 seconds from CSV upload
No explanation for why an account is flagged	Gemini AI writes plain-English reason per flagged account
No visual overview of fraud connections	Interactive network graph shows the full fraud picture

11. Future Scope

Enhancement	Description
UPI Fraud Detection	Extend model to detect suspicious UPI transaction patterns in real time
Real-Time Transaction Blocking	Integrate with bank APIs to auto-block high-risk transactions instantly
Cross-Bank Fraud Network Discovery	Connect data across multiple banks to detect fraud networks spanning institutions
National Fraud Intelligence Feed	Ingest RBI / I4C cyber fraud alerts for enriched risk scoring
Advanced Behavioral Analytics	Build per-customer behavioral profiles for more accurate anomaly detection